Expression of the biochemical defect of methionine dependence in fresh patient tumors in primary histoculture.

. . . . .

Methionine dependence is a metabolic defect that occurs in many human tumor cell lines but not normal in unestablished cell strains. Methionine-dependent tumor cell lines are unable to proliferate and arrest in the late S/G2 phase of the cell cycle when methionine is replaced by its immediate precursor homocysteine in the culture medium (MET-HCY+ medium). However, it is not known whether methionine dependence occurs in fresh patient tumors as it does in cell lines. In order to determine whether methionine dependence occurs in fresh patient tumors as well as whether methionine dependence occurs in fresh patient tumors as well as in cell lines we took advantage of the technique of sponge-gel-supported histoculture to grow tumors directly from surgery. We then measured nuclear DNA content by image analysis to determine the cell cycle position in MET-HCY+ compared to MET+HCY- medium in 21 human patient tumors. Human tumor cell lines found to be methionine dependent by cell count were used as positive controls and were found to have marked reduction of cells in G1 compared to total cells in the cell cycle in MET-HCY+ medium with respect to the G1: total cell ratio in MET+HCY- medium. Therefore late cell cycle arrest was used as a marker of methionine dependence for histocultured patient tumors. We found that 5 human tumors of 21, including tumors of the colon, breast, ovary, prostate, and a melanoma, were methionine dependent based on cell cycle analysis. These data on fresh human tumors indicate that methionine dependence may frequently occur in the cancer patient population. Implications for potential therapy based on methionine dependence are discussed.